In [1]:
```
#importing the libraries
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras import datasets, layers, models
```

```
WARNING: All log messages before absl::InitializeLog() is called are writt
en to STDERR
I0000 00:00:1793420102.208093    8683 port.cc:153] oneDNN custom operation
s are on. You may see slightly different numerical results due to floatin
g-point round-off errors from different computation orders. To turn them o
ff, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1793420102.213140    8683 cudart_stub.cc:31] Could not find cu
da drivers on your machine, GPU will not be used.
I0000 00:00:1793420102.661302    8683 cpu_feature_guard.cc:227] This Tenso
rFlow binary is optimized to use available CPU instructions in performanc
e-critical operations.
To enable the following instructions: AVX2 AVX_VNNI FMA, in other operatio
ns, rebuild TensorFlow with the appropriate compiler flags.
/usr/lib/python3/dist-packages/scipy/__init__.py:146: UserWarning: A NumPy
version >=1.17.3 and <1.25.0 is required for this version of SciPy (detect
ed version 1.26.4
  warnings.warn(f"A NumPy version >={np_minversion} and <{np_maxversion}"
WARNING: All log messages before absl::InitializeLog() is called are writt
en to STDERR
I0000 00:00:1793420103.848591    8683 port.cc:153] oneDNN custom operation
s are on. You may see slightly different numerical results due to floatin
g-point round-off errors from different computation orders. To turn them o
ff, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1793420103.850999    8683 cudart_stub.cc:31] Could not find cu
da drivers on your machine, GPU will not be used.
```

In [2]:
```
#grabbing CIFAR10 dataset
(train_images, train_labels), (test_images, test_labels) = datasets.cifar
train_images, test_images = train_images / 255.0, test_images / 255.0
```

In [3]:
```
#showing images of mentioned categories
class_names = ['airplane', 'automobile', 'bird', 'cat', 'deer','dog', 'fr

plt.figure(figsize=(10,10))
for i in range(10):
    plt.subplot(5,5,i+1)
    plt.xticks([])
    plt.yticks([])
    plt.grid(False)
    plt.imshow(train_images[i])
    plt.xlabel(class_names[train_labels[i][0]])
plt.show()#model compilation
model.compile(optimizer='adam',loss=tf.keras.losses.SparseCategoricalCros
epochs = 1
h = model.fit(train_images, train_labels, epochs=epochs, validation_data=
```




```
---------------------------------------------------------------------------
-
NameError                                 Traceback (most recent call las
t)
/tmp/ipykernel_8683/4138620027.py in <module>
     11     plt.xlabel(class_names[train_labels[i][0]])
     12 plt.show()#model compilation
---> 13 model.compile(optimizer='adam',loss=tf.keras.losses.SparseCategori
calCrossentropy(from_logits=True),metrics=['accuracy'])
     14 epochs = 1
     15 h = model.fit(train_images, train_labels, epochs=epochs, validatio
n_data=(test_images, test_labels))

NameError: name 'model' is not defined
```

In [4]:
```
#building CNN model
model = models.Sequential()
model.add(layers.Conv2D(32, (3, 3), activation='relu', input_shape=(32, 3
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(64, (3, 3), activation='relu'))
model.add(layers.MaxPooling2D((2, 2)))
model.add(layers.Conv2D(64, (3, 3), activation='relu'))
model.add(layers.Flatten())
model.add(layers.Dense(64, activation='relu'))
model.add(layers.Dense(10))

model.summary()
```

```
/home/hardware/.local/lib/python3.10/site-packages/keras/src/layers/convol
utional/base_conv.py:113: UserWarning: Do not pass an `input_shape`/`input
_dim` argument to a layer. When using Sequential models, prefer using an `
Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
E0000 00:00:1793420118.323778    8683 cuda_platform.cc:52] failed call to
cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 30, 30, 32)	896
max_pooling2d (MaxPooling2D)	(None, 15, 15, 32)	0
conv2d_1 (Conv2D)	(None, 13, 13, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 6, 6, 64)	0
conv2d_2 (Conv2D)	(None, 4, 4, 64)	36,928
flatten (Flatten)	(None, 1024)	0
dense (Dense)	(None, 64)	65,600
dense_1 (Dense)	(None, 10)	650

Total params: 122,570 (478.79 KB)

Trainable params: 122,570 (478.79 KB)

Non-trainable params: 0 (0.00 B)

In [ ]:
```
#model compilation
model.compile(optimizer='adam',loss=tf.keras.losses.SparseCategoricalCros
epochs = 1
h = model.fit(train_images, train_labels, epochs=epochs, validation_data=
```

In [ ]: